

An Automatic Micro-calcification Detection Method on Mammograms Based on Fractal Dimension

Shen-Chuan Tai^{1*}, Zih-Siou Chen^{1*}, Chin-Peng Lin^{1*}, and Dr. Li-li Cheng²

¹Institute of Computer and Communication Engineering, Department of Electrical Engineering, National Cheng Kung University, Tainan 701, Taiwan, R.O.C.

{sctai, czs99d, lcp100m}@dcmc.ee.ncku.edu.tw

²National Cheng Kung University Hospital, 138, Sheng Li Road, Tainan 704, Taiwan, R.O.C.

lili_112005@yahoo.com.tw

Abstract

In this paper, an automatic micro-calcification detection method on mammograms is proposed. The main scheme can be briefly divided to two stages: preprocessing stage and regions of interest (ROIs) extraction stage. In the first stage, the smoothed sobel edge map is multiplied by the sharpened image processed from the unsharp masking. This can enhance the high frequency details including the micro-calcifications (MCs) in the original image. On the contrary, other high frequency details, i.e. noises, are reduced. In the second stage, the breast region is divided to several non-overlapped blocks. Then the fractal dimension and the standard deviation are both applied to describe the texture characteristics in each block. Blocks with high fractal dimension and standard deviation are more likely to be MC regions. Eventually, a parameter related to the detection precision is user-defined to automatically detect the MC regions. Experiments show that the detection results can reduce the radiologists' workload.

1 Introduction

For years, cancer causes a high mortality rate for people and it has been gradually trending upward, especially in the developed and under-developed countries. In Taiwan, it was estimated that cancer ranked first among all the causes of death in 2011, representing 28% of total cases (29% in men and 26.4% in women) [1].

Breast cancer is one of the leading causes of cancer death in women but until now, there have not been effective ways to prevent from its occurrence yet. However, it is evidence that early

detection accompanied with early treatment can increase the chance of survival for the patients [2].

One of the significant early sign of breast abnormalities is the appearance of micro-calcification(MC) clusters, and they appear in 30-50% of the mammographic diagnosed cases [3][4]. MCs are tiny specks of calcium deposits (with an average diameter of 0.3mm for individual MCs) that can be scattered throughout the mammary gland or gather in clusters [5]. MCs vary with their sizes, shapes and distributions. Round or oval shaped MCs uniform in size tend to be associated with a benign process and small, irregular, polymorphic, branching MCs heterogeneous in size and morphology to a malignant process [6]. Figure 1 shows the examples, where (a) and (b) belong to benign findings and (c) is a malignant finding.

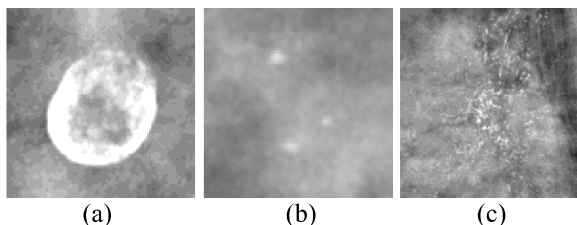


Figure 1. Benign and malignant MC cases

At the present day, there are lots of imaging modalities, where mammogram and ultrasound imaging are well-known. Nevertheless, MCs are invisible in ultrasound images. As a matter of fact, when it comes to MC detection, mammography performs better sensitivity as compared to its competitors. However, manual reading of mammograms is time-consuming and tiring. Even experienced radiologists may sometimes misdiagnose due to visual fatigue during long-time concentration. In fact, 10-30% of breast lesions are missed during routine screening [7][8]. Thus, computer aided diagnosis (CAD) systems are brought to assist specialists in diagnosis as a second opinion. Studies demonstrated that average sensitivity of radiologists can be increased by 10%

*The authors are with Data Compression & Multimedia Communication Laboratory.

with the aid of CAD systems [7].

For many years, CAD systems have drawn much attention to researchers, several techniques and algorithms have been proposed subsequently. Leiner et al. [9] used two-dimensional discrete wavelet transform and image enhancement techniques for removing noise as well as to obtain a better contrast. Bhateja et al. [10] proposed a novel framework for edge detection of microcalcifications using a combination of non-linear enhancement operator and morphological filter. Mohanalin et al. [11] proposed a fuzzy based approach for denser mammographic images. Wei et al. [12] proposed an automatic detection system by implementing wavelet coefficient to extract suspicious regions (i.e. ROI) and then using DOG filter with adaptive thresholds on those regions to segment MCs. Karahaliou et al. [13] investigated texture properties of the tissue surrounding MC clusters. Several features describing the tissue surrounding MCs were fed to a probabilistic neural network as classifier. Results suggested that this method can contribute to CAD of breast cancer. Tsai et al. [14] used wavelet layers and Renyi's information theory to separate the suspicious MC regions. Then, features related to those regions were transformed by principal component analysis to a compact and efficient expression by linear combination method. In the last, back-propagation neural network played an important role in the classification stage.

Although advances have been made much effort in the field of mammographic CAD systems, MCs detection is still a challenging task. For one thing, mammograms are often poor in contrast. Secondly, MCs themselves have fuzzy nature and are hard to be distinguished from their surroundings, especially in the dense breast parenchyma [7]. Hence, an accurate detection of MCs is highly desirable. An example of dense mammogram of MCs is shown in figure 2. Notice that the black circles denote the MC clusters.

In this paper, an automatic MC detection algorithm is presented which can be roughly divided into two stages: preprocessing stage and regions of interest (ROIs) extraction stage. In the first stage, the smoothed sobel edge map is multiplied by the sharpened image processed from the unsharp masking. This can enhance the high frequency details including the microcalcifications (MCs) in the original image. On the contrary, other high frequency details, i.e. noises, are reduced. In the second stage, the breast region is divided to several non-overlapped blocks. Then the fractal dimension and the standard deviation are both applied to describe the texture characteristics in each block. Blocks with high

fractal dimension and standard deviation are more likely to be MC regions. Eventually, a parameter related to the detection precision is user-defined to automatically detect the MC regions. That is to say, the parameter with higher value can detect tinier MCs, but increase the false positives oppositely.

The paper is organized as follows: in the section 2, the mammographic database is introduced; the section 3 describes the proposed method and the section 4 demonstrates the experimental results; finally, some conclusions remain in the section 5.

2 Mammogram Database

The mammograms are collected at the National Cheng Kung University Hospital, Tainan, Taiwan. Each mammogram is stored as DICOM format with size 3328*2560, 12 bit depth, and 0.07*0.07 mm² per pixel. There are total 158 mammograms containing 585 MC clusters based on radiologists' opinions. In addition, all images are adjusted to 8-bit in order to be displayed.

There are two standard mammographic views: the craniocaudal (CC) view (shown in figure 3(a)) and the mediolateral (MLO) view (shown in figure 3(b)). The CC view images the breast from above but MLO view is taken from an oblique or angled view. The MLO view is the most important projection as it allows to depict most breast tissue. For example, the MLO view depicts the pectoral muscle but the CC view does not. Furthermore, it is possible that MCs occur on the pectoral muscle region, as shown in figure 4. Thus, the proposed method is aimed at the MLO view. Notice that the black line in figure 4 denotes the pectoral muscle margin.

3 Proposed Method

The main scheme of the proposed method is shown in figure 5.

3.1 Preprocessing Stage

Achievement to emphasize MCs without over-enhancement is highly expected. In this stage, the breast region is firstly extracted from the maximum connected component labeling [15], and thus the labels and markers are removed simultaneously. Notice that the pectoral muscle region should not be removed due to possible occurrence of MCs. Subsequently, the unsharp masking is used to obtain a sharpened image, and the sobel operator together with an average filter produce a smoothed sobel edge map.

3.1.1 Unsharp Masking

Since MCs are the high-frequency component in an image, the unsharp masking is applied to the preprocessed images. Unsharp masking is a simple sharpening technique which enhances the high frequency information and emphasizes the edge and details. However, it is very sensitive to noise [16].

In this paper, the unsharp masking is implemented by adding a laplacian filtered image to the input image, as shown in figure 6. Where $I(x,y)$, $O(x,y)$ and $P(x,y)$ denote the input image, the output image and the high frequency details of the input image enhanced by the laplacian filter respectively. Briefly, $P(x,y) = 4*I(x,y) - I(x-1,y) - I(x+1,y) - I(x,y-1) - I(x,y+1)$. Accordingly, MCs along with noise are enhanced simultaneously.

3.1.2 Smoothed Sobel Edge Map

The sobel operator [15] is a popular edge detection technique due to the fact that the sobel masks have better noise-suppression characteristics and are easy to implement. It is a discrete differentiation operator, computing an approximation of the gradient of the image intensity. After the sobel operator is applied, the sobel edge map is obtained. Then a 5*5 average filter is applied on the sobel edge map to reduce the noise on the map.

3.1.3 Multiplication

For the sake of noise reduction and significant detail reservation, the unsharp masking image is multiplied by the smoothed sobel edge map [15]. From this, the enhanced details by the unsharp masking are reserved with the aid of the sobel edges. On the other hand, noises in the smooth regions are suppressed owing to the average filter used before.

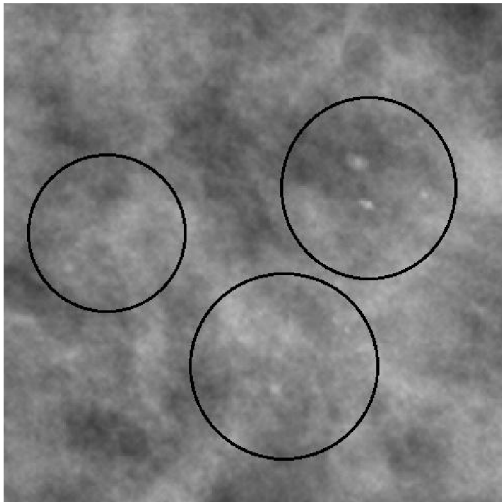


Figure 2. An example of MC clusters in dense mammogram

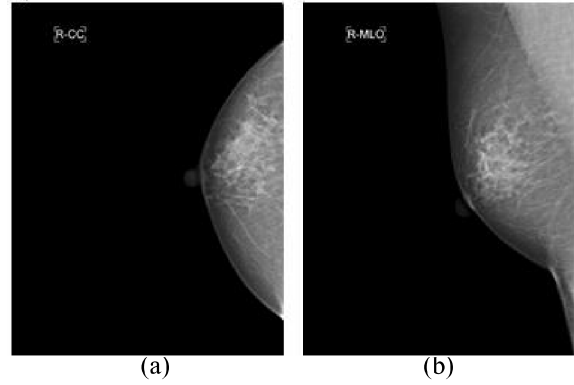


Figure 3. The CC and MLO view

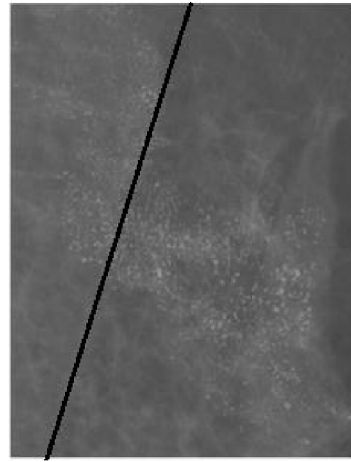


Figure 4. The MCs on the pectoral muscle region

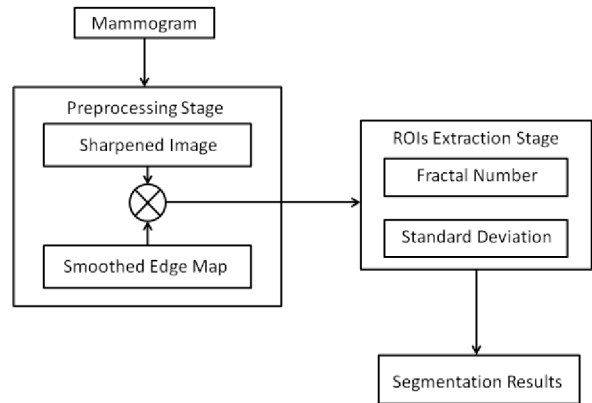


Figure 5. Main Scheme of the proposed method

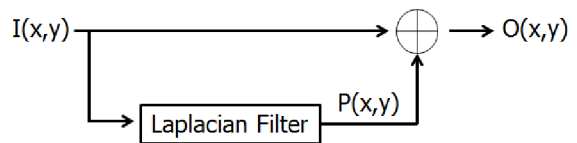


Figure 6. Flow chart of unsharp masking

3.2 Regions of Interest (ROIs) Extraction Stage

In this stage, the preprocessed breast region is divided to a series of non-overlapping sub-image blocks. The sub-image block size chosen has an influence on the MC detection precision. In other words, larger blocks enable to quickly screen mammograms to find large calcified areas, while the smaller blocks are used to detect small and isolated MCs [17]. MC size typically ranges between 0.05mm and 1mm, which is equivalent to 7-143 pixel length for individual MC. Thus, block size of 8*8 is appropriate in an attempt to achieve high precision at the cost of computation time.

In each block, the fractal dimension (FD) and the standard deviation (STD) are calculated to identify if there are MCs in the block. FD measures the texture characteristics but is not robust to too tiny or ambiguous MCs. Thus, STD is considered as compensation for its advantage to calculate the variability of data sets. Then several candidate regions are retained as long as the blocks are with FD and STD higher than a threshold C. Subsequently, the 8-connectivity labeling is used to connect the candidate regions together to be ROIs. The ROIs represent the final MC detection results.

There have been many methods to estimate the fractal dimension, the blanket method proposed by Peleg is used [17][18]. The computational formula in [17] is as follows:

$$FD = 2 - \frac{(\log A_r - \log K)}{\log r}$$

where, A_r is the fractal surface area evaluated by the neighborhood of a pixel when a given preset scaling factor r (see [17] for detail description) and K represents the surface area as the scaling factor is equal to 1.

4 Experiments

The brief process is presented as follows. Notice that all figures are adjusted to 8 bit for display.

- Step 1: The breast region is extracted from the maximum connected component, as shown in figure 7.

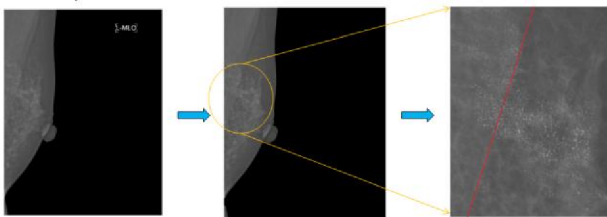


Figure 7.

- Step 2.1: The laplacian filter is used on the breast region to obtain a sharpened image, as shown in figure 8.

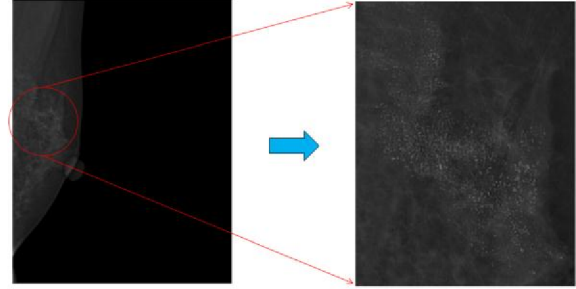


Figure 8.

- Step 2.2: To suppress the enhanced noises caused by laplacian filter, the sobel operator and the average filter are used together, as shown in figure 9.

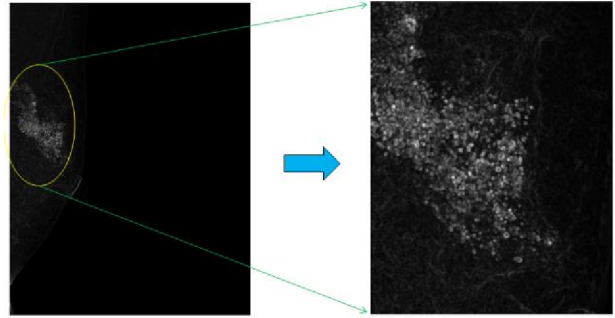


Figure 9.

- Step 3: The sharpened image from step 2.1 is multiplied by the smoothed sobel edge map from step 2.2, as shown in figure 10.



Figure 10.

- Step 4: FD and STD are calculated and the results are shown in figure 11 with adjusted

image contrast. Notice that the irregular contour representing MC cluster is drawn by radiologist and the rectangle lines present the detection results.

5 Conclusion

In this paper, an automatic micro-calcification detection method on mammograms is proposed. By multiplying the sharpened image by the smoothed sobel edge map, significant high frequency details are enhanced without increasing noise. Then texture descriptor FD along with compensation of STD, are introduced to get the detection results.

Figure 11 demonstrates the result of a case with C equivalent to 200. If C is adjusted higher, the number of detected blocks increases, and thus less MC clusters are missed. Nevertheless, this increases the false positives at the same time. As a matter of fact, as C is equivalent to 200, the true positives are 571 out of totally 585 true MC clusters drawn by radiologists. While C is adjusted to 1000, the true positives rise to 585. This provides an user-defined platform and the detection results help the radiologists reduce their workload.

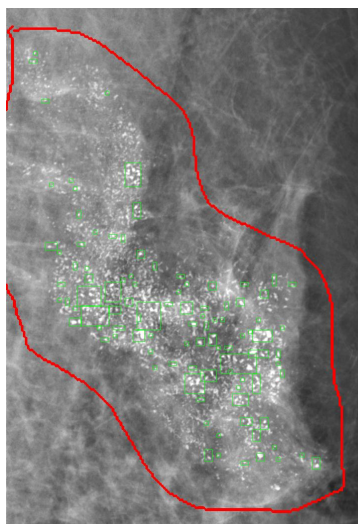


Figure 11. Detection result with C equivalent to 200.

Acknowledgement

Sincerely thank Miss Chia-Ying Lin and Miss Hsing-Fang Li for helping us collect the mammograms and provide the ground truth of them together with related medical knowledge.

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